

Finite-Sample Exclusion of Spectral Gaps from Gaussian Correlator Sketches

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<https://github.com/lluiseiriksson/finite-sample-spectral-certificates>

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Abstract

Spectral gaps are often inferred from noisy Euclidean correlators by fitting a small number of exponentials or solving a generalized eigenvalue problem. Such point estimates do not by themselves control the error incurred when the spectral filter is selected from the same data. We give a finite-sample test that instead asks whether an entire simultaneous confidence set is compatible with a proposed upper edge of the visible transfer spectrum. For Gaussian sketches of a block correlator, an extreme-singular-value inequality yields a Loewner confidence band for the unknown joint Gram matrix. Intersecting this band with a truncated Hausdorff localizer is one semidefinite feasibility problem. We derive its semidefinite alternative: every rejection is accompanied by a covariance-aware matrix witness, a negative separation margin, and checkable primal-independent residuals. The test has type-I error at most α at fixed truncation order and requires neither known covariance nor a Gaussian approximation to the quadratic moment features. Simulations in the interacting axial next-nearest-neighbour Ising chain compare the certificate to covariance-known ellipsoids, plug-in eigenvectors, fixed Chebyshev filters, and Ritz estimates. We also document a negative result: Hotelling calibration can severely undercover the non-Gaussian quadratic features at moderate sample size. The method certifies exclusion of overly large *visible* gaps; it does not turn non-rejection into a lower bound on a Hamiltonian mass gap.

1 Introduction

Let H be a Hamiltonian with ground energy E_0 , choose a Euclidean step $\tau > 0$, and write $T = \exp[-\tau(H - E_0)]$. Matrix-valued correlator moments have the form

$$B_k = \Psi^* T^k \Psi, \quad k = 0, 1, \dots, \quad (1)$$

where the columns of Ψ are ground-state-subtracted probe states. The largest point in the support visible to these probes is $\rho_{\text{vis}} = e^{-\tau \Delta_{\text{vis}}}$. If a probe misses the first excitation then Δ_{vis} may exceed the Hamiltonian gap; this visibility qualification is unavoidable.

Recent work has developed semidefinite bootstraps for exact Euclidean correlators [1], convergent lower-gap certificates for qubit Hamiltonians [2], and convergent certified upper bounds on the bulk gap [3]. Our question is complementary:

Given finitely many correlated random observations of a block correlator, can one reject a proposed visible gap with simultaneous, finite-sample error control while choosing the separating filter adaptively?

The distinction matters. A negative eigenvalue of a noisy localizer is a data-dependent observation, not yet a confidence statement. Entrywise error bars ignore correlation; a filter trained on the same sample invalidates a pointwise interval; and a Gaussian approximation to quadratic observables may be inaccurate precisely in the moderate- n regime where certification is most valuable.

Our contribution is a statistical-conic layer that can be placed in front of a truncated moment localizer. In the Gaussian-sketch model the primitive sketch vector is Gaussian, but the reported block moments are quadratic and Wishart-like. We construct a confidence set for the primitive covariance itself, then test its compatibility with the spectral null. The resulting test is nonasymptotic and covariance-free. Its dual supplies an explicit rejection witness rather than asking the reader to trust an “infeasible” solver label.

Scope. At finite moment degree, localizer positivity is a necessary condition for a proposed upper spectral edge. Its violation rigorously excludes that edge, but non-violation does not prove a gap. Consequently our output is a certified upper bound on a visible gap (or a rejection of an overly large claimed gap), not a proof of a positive mass gap. This boundary also distinguishes the present result from Hamiltonian-algebra hierarchies that can certify lower bounds [2].

2 Block moments and the upper-edge localizer

Assume for clarity that H and the probes admit a real representation; the complex proper-Gaussian case has different fourth-moment constants and is not claimed here. The spectral theorem gives a positive-semidefinite matrix-valued measure μ on $[0, 1]$ such that

$$B_k = \int_{[0,1]} x^k d\mu(x). \quad (2)$$

For a proposed edge $\theta \in (0, 1]$ and degree N , define

$$\mathcal{L}_{N,\theta}(B) = [\theta B_{i+j} - B_{i+j+1}]_{i,j=0}^N. \quad (3)$$

If $\text{supp } \mu \subseteq [0, \theta]$, then for every vector polynomial $p(x) = \sum_{i=0}^N x^i v_i$,

$$\sum_{i,j} v_i^* (\theta B_{i+j} - B_{i+j+1}) v_j = \int (\theta - x) p(x)^* d\mu(x) p(x) \geq 0. \quad (4)$$

Thus $\mathcal{L}_{N,\theta}(B) \succeq 0$ is a necessary finite-window null. A violation says that the probes see spectral weight above θ , hence

$$\Delta_{\text{vis}} < -\tau^{-1} \log \theta. \quad (5)$$

This elementary localizer is not presented as novel; truncated moment and localization techniques have a long history [4, 5]. The new issue below is how to test it after observing a correlated random Gram matrix.

3 Gaussian sketches and a covariance confidence band

Put $u_{k,a} = T^{k/2} \psi_a$ for time index $k = 0, \dots, 2N + 1$ and channel $a = 1, \dots, c$. Draw independent standard real Gaussian vectors g_r and observe

$$z_{r,k,a} = \langle g_r, u_{k,a} \rangle. \quad (6)$$

After stacking (k, a) into one index, $z_r \sim \mathcal{N}(0, K)$ with

$$K_{(k,a),(\ell,b)} = \langle u_{k,a}, u_{\ell,b} \rangle = B_{(k+\ell)/2}[a, b]. \quad (7)$$

In particular, the k th diagonal time block of K is B_k . Although $z_{r,k} z_{r,k}^\top$ is an unbiased rank-one estimator of B_k , these quadratic features are not jointly Gaussian.

Let

$$S = \frac{1}{n} \sum_{r=1}^n z_r z_r^\top, \quad d = (2N + 2)c. \quad (8)$$

The following confidence set acts on all times and channels simultaneously.

Theorem 1 (Wishart–Loewner confidence band). *Let $z_1, \dots, z_n$ be independent $\mathcal{N}(0, K)$ vectors in $\mathbb{R}^d$, where $K \succeq 0$ is unknown and the zero mean is known. For $0 < \alpha < 1$, set*

$$\eta = \frac{\sqrt{d} + \sqrt{2\log(2/\alpha)}}{\sqrt{n}} < 1, \quad a = (1 + \eta)^{-2}, \quad b = (1 - \eta)^{-2}. \quad (9)$$

Then

$$\mathbb{P}\{aS \preceq K \preceq bS\} \geq 1 - \alpha. \quad (10)$$

The statement remains valid for singular K .

Proof. Write the $n \times d$ data matrix as $Z = GK^{1/2}$, with G standard Gaussian. The Gaussian extreme-singular-value inequality gives, with failure probability at most $2e^{-t^2/2}$,

$$\sqrt{n} - \sqrt{d} - t \leq s_{\min}(G) \leq s_{\max}(G) \leq \sqrt{n} + \sqrt{d} + t.$$

With $t = \sqrt{2\log(2/\alpha)}$, division by $\sqrt{n}$, squaring, and congruence by $K^{1/2}$ yield $(1 - \eta)^2 K \preceq S \preceq (1 + \eta)^2 K$. Rearrangement gives (10). Congruence does not require invertibility of K , so the same argument covers the singular case. We use the ambient d rather than an unknown effective rank. See [6, 7] for the nonasymptotic singular-value bounds. $\square$

Remark 2 (Known mean is substantive). *If a sample mean is first estimated and subtracted, the degrees of freedom and constants change. The present theorem applies to the centered-by-construction sketches in (6); it must not be copied unchanged to an unknown-mean observation model.*

4 Exact compatibility test and semidefinite alternative

Let $P(K) = (K_{kk})_{k=0}^{2N+1}$ extract the diagonal time blocks. Define the random confidence set

$$\mathcal{C}_\alpha(S) = \{Q \in \mathbb{S}^d : aS \preceq Q \preceq bS\}. \quad (11)$$

We reject the proposed edge θ precisely when

$$\text{there is no } Q \in \mathcal{C}_\alpha(S) \text{ with } \mathcal{L}_{N,\theta}(P(Q)) \succeq 0. \quad (12)$$

This is one SDP feasibility problem.

Corollary 3 (Finite-sample validity). *Under the null $\mathcal{L}_{N,\theta}(P(K)) \succeq 0$, the probability that (12) rejects is at most α .*

Proof. On the event in theorem 1, the true K is feasible in (12); rejection is therefore impossible. The rejection event is contained in the confidence-set failure event. $\square$

No sample splitting is needed: arbitrary adaptive optimization inside the confidence set preserves this argument. This is the central statistical advantage over selecting a polynomial direction from S and attaching an unadjusted pointwise error bar.

For numerical auditability, let D be any fixed positive diagonal scaling computed from S , put $Q = D\tilde{Q}D$, and denote the scaled interval by $A \preceq \tilde{Q} \preceq B$. Let $\mathcal{L}_D$ be the resulting affine linear map.

Proposition 4 (Semidefinite alternative). *Assume $S \succ 0$. The feasibility problem (12) is infeasible if and only if there exist $Y_A, Y_B, Y_L \succeq 0$ such that*

$$Y_A - Y_B + \mathcal{L}_D^*(Y_L) = 0, \quad (13)$$

$$-\langle Y_A, A \rangle + \langle Y_B, B \rangle < 0. \quad (14)$$

The multipliers may be normalized by $\text{tr } Y_A + \text{tr } Y_B + \text{tr } Y_L = 1$.

Proof. Pair the three primal inequalities $\tilde{Q} - A \succeq 0$, $B - \tilde{Q} \succeq 0$, and $\mathcal{L}_D(\tilde{Q}) \succeq 0$ with the three PSD multipliers. Equation (13) cancels every term containing $\tilde{Q}$; primal feasibility would then force the constant in (14) to be nonnegative, proving one direction. Conversely, the interval is compact and has the strict midpoint $(A + B)/2$ when $S \succ 0$. Strong conic separation therefore supplies the stated multipliers whenever its intersection with the inverse image of the PSD localizer cone is empty. Homogeneity permits trace normalization. If S is singular, the same claim requires first restricting all matrices to $\text{ran } S$; we do not silently replace this reduction by a numerical ridge. $\square$

The implementation minimizes the dual margin under trace normalization. A reported rejection stores the margin, the Frobenius residual of (13), normalization error, and the smallest eigenvalue of each numerical multiplier. Interval or rational outward rounding remains the last step from an auditable floating-point witness to a machine-checkable one.

5 Why covariance awareness changes the witness

The noiseless localizer test uses a minimum eigenvector. Under simultaneous uncertainty, that direction need not be optimal. For comparison, if a moment vector y has known mean covariance C/n and an ellipsoidal radius q , a trace-one PSD witness X has population signal-to-noise ratio

$$\kappa(X) = \frac{[-\text{tr}(\mathcal{L}(y)X)]_+}{\sqrt{\ell(X)^\top C \ell(X)}}, \quad \ell(X)_k = \text{tr}(L_k X). \quad (15)$$

The robust upper bound becomes negative exactly when $\sqrt{n} \kappa(X) > q$. Optimizing X therefore trades localizer negativity against covariance sensitivity, rather than simply minimizing the observed eigenvalue.

Mixed witnesses can be strictly better in principle. For the synthetic family $M = -\varepsilon I_r$ with isotropic Frobenius uncertainty, every rank-one projector has uncertainty norm one, whereas $X = I_r/r$ has norm $r^{-1/2}$. The sample threshold differs by a factor r . In our ANNNI pilots, however, many optimized ellipsoidal witnesses above threshold are numerically rank one. The observed advantage there is primarily *covariance-aware direction selection* relative to the plug-in minimum eigenvector, not evidence that a high-rank mixture is always required.

6 Interacting ANNNI experiment

We use the periodic axial next-nearest-neighbour Ising chain

$$H = -J_1 \sum_i Z_i Z_{i+1} - J_2 \sum_i Z_i Z_{i+2} - h_x \sum_i X_i \quad (16)$$

with $J_1 = 1$, $J_2 = 0.37$, $h_x = 2.2$, and $\tau = 0.2$. This frustrated, nonintegrable finite chain is not being used to claim a new phase boundary; it is a controlled interacting test bed where exact sparse diagonalization and Krylov propagation provide ground truth. Ground-state-subtracted central-site X and Z probes form a two-channel correlator. The global spin-flip parity separates their visible sectors.

At degree $N = 2$, the primitive joint sketch dimension is $d = 12$, the flattened quadratic moment dimension is 18, and the localizer has size 6×6 . Lengths $L = 6, 8, 10, 12, 14, 16$ are tested at the exact visible edge and at gaps inflated by 15% and 35%. For each cell we generate independent sketch ensembles at $n = 500, 1000, 2000, 5000$. The production seed, full raw summaries, and SHA-256 hashes are stored in the repository.

6.1 Baselines and controls

We compare four logically different objects.

1. The Wishart–Loewner dual certificate of Proposition 4, with unknown covariance and finite-sample level control.
2. A covariance-known multivariate-Chebyshev ellipsoid, useful as a controlled oracle reference but not a deployable unknown-covariance method.
3. A plug-in minimum-eigenvector and a fixed Chebyshev polynomial filter. These expose the value of covariance-aware direction selection.
4. A generalized Ritz/GEVP edge estimate. It is a point estimator, not a confidence certificate, and is reported separately rather than granted the same logical status.

The known-covariance pilot showed a sharp sample transition: for the 35% inflated gap, the covariance-aware robust test reached essentially full power near $n = 1000$ across $L \leq 14$, while the plug-in eigenvector was usually still silent until $n = 5000$. At the true edge the robust methods made no rejection in the medium grid. These observations motivated the covariance-unknown production campaign; final rates and binomial intervals will replace this paragraph after the full artifact is frozen.

6.2 Negative calibration result

The quadratic sketch features obey Isserlis’ covariance formula but are Wishart-like rather than multivariate Gaussian. At $L = 12$, $N = 2$, and only $n = 120$ sketches, a nominal 95% Hotelling ellipsoid covered the exact moment vector in 83.4% of 500 simulated ensembles. Its apparent 97.4% detection rate for an inflated gap was therefore invalid as a 95% certificate. Coverage approached nominal only slowly with n . This failed route is retained in the artifact as a warning against importing an exact Gaussian-mean theorem into a quadratic-feature problem.

7 Numerical verification protocol

Ill-conditioned Euclidean moment matrices make raw primal status unreliable. In a first primal–dual Wishart replay, two of five inflated-gap samples were labeled `optimal_inaccurate` by a primal solver even though their dual margins were below -4×10^{-4} . In a separate ten-ensemble $L = 8$ smoke at $n = 500$, null dual objectives lay between 1.8×10^{-10} and 6.5×10^{-10} , while 35%-inflated-gap objectives lay between -8.2×10^{-4} and -4.1×10^{-4} . The maximum stationarity residual was 2.3×10^{-9} . The first production pass conservatively thresholded the raw dual margin at -10^{-7} and stored all residual diagnostics. The final pass uses the feasibility repair described next.

Feasibility repair. Let floating-point multipliers first be projected onto the PSD cone, and let $R = Y_A - Y_B + \mathcal{L}_D^*(Y_L) = R_+ - R_-$ be their remaining stationarity residual. The replacements

$$\hat{Y}_A = Y_A + R_-, \quad \hat{Y}_B = Y_B + R_+ \tag{17}$$

remain PSD and satisfy stationarity exactly, up to the arithmetic used for the eigendecomposition. Trace renormalization preserves this equality. In a 20-ensemble repaired smoke, the maximum stationarity residual fell below 8×10^{-17} and normalization error below 4×10^{-16} ; null objectives were positive, whereas all 15% and 35% inflated-gap objectives were negative. The production threshold is -10^{-9} on the *repaired* objective, leaving an additional buffer above double-precision roundoff.

This gap between scientific margin and numerical residual is encouraging but does not itself constitute interval arithmetic. The release protocol is:

1. solve in floating point after diagonal congruence scaling;
2. independently rebuild the adjoint and residual from exported matrices;
3. project only for diagnostics, never to manufacture a better objective;
4. outward-round selected repaired witnesses and verify the resulting inequalities with interval or rational arithmetic;
5. rerun a subset with a second conic solver.

8 Limitations and interpretation

Visible versus Hamiltonian gap. The probes may have zero overlap with the first excitation. Multiple symmetry channels reduce but do not remove this risk. Every result is consequently labeled Δ_{vis} unless a separate overlap theorem is available.

Upper versus lower bounds. Rejecting support in $[0, \theta]$ proves that a claimed gap is too large. A finite localizer’s non-rejection is not a proof that the true support lies below θ . Our method therefore complements, rather than replaces, lower-gap Hamiltonian certificates [2].

Acquisition model. The exact confidence band uses independent, known-zero-mean Gaussian sketch coordinates. Classical stochastic estimators or randomized numerical linear algebra can realize this model directly; a laboratory implementation must state how those coordinates are obtained and account for state-preparation and measurement noise. Markov-chain autocorrelation, non-Gaussian primitive sketches, and unknown centering require different concentration results.

Finite volume. The ANNNI study validates statistical behavior at finite L . It neither establishes a thermodynamic gap nor competes with the convergent bulk hierarchy of Xu et al. [3]. A future combination of noisy-data confidence sets with thermodynamic operator-algebra hierarchies is an open direction.

9 Conclusion

The main result is a finite-sample bridge from random correlator sketches to an auditable spectral-edge exclusion. A single Loewner band absorbs all time-channel correlations and all adaptive witness selection; a semidefinite alternative turns rejection into a replayable matrix inequality. The approach is intentionally one-sided. It can falsify an overly large visible gap with controlled type-I error, but it does not infer a mass gap from silence. The interacting-chain campaign is designed to quantify when that rigorous statement is statistically useful and when conventional point estimates are merely more optimistic.

Reproducibility. Code, theorem audit notes, raw pilot outputs, the Colab production notebook, and the evolving manuscript are available at <https://github.com/lluiseriksson/finite-sample-spectral-certificates>. The research branch and draft pull request remain open until the production artifact and interval replay are frozen.

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